Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 30 epochs with a batch size of 128. The learning rate was set to 0.01 with the Adam optimizer.

2 Results

Figure 1 shows the test accuracy during training.

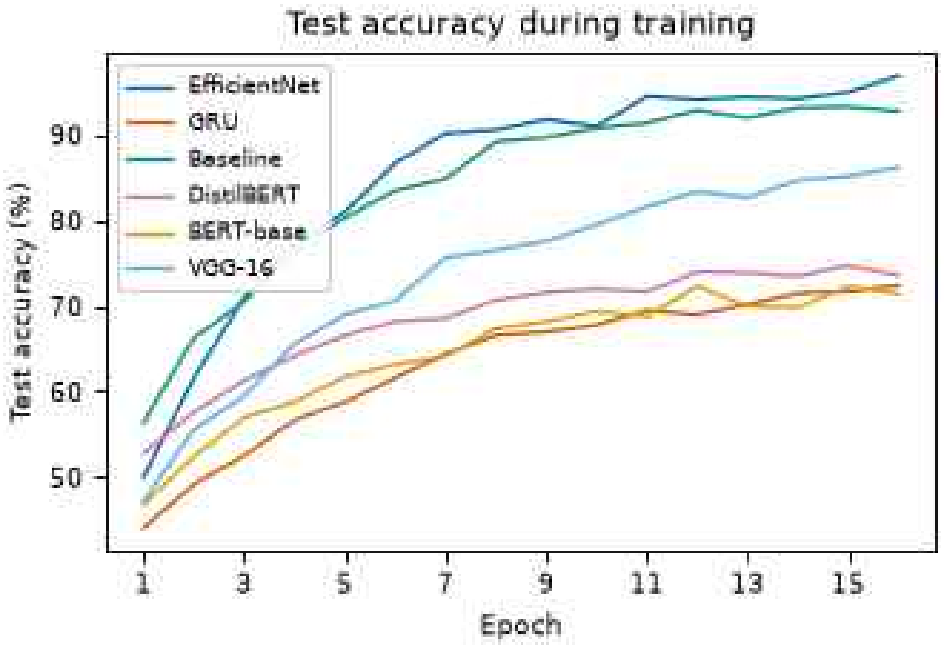


Figure 1: Test accuracy during training for the models in this experiment.

Epoch 1 sees DistilBERT post a test accuracy of 52.8% (Figure 1). At epoch 3, EfficientNet leads GRU by 19.7 points (Figure 1). From epoch 2 to epoch 14, the test accuracy of GRU improves by a relative 51.2%. By epoch 9, the test accuracy of DistilBERT stands at 65.8% (Figure 1).

Figure 2 shows the training loss.

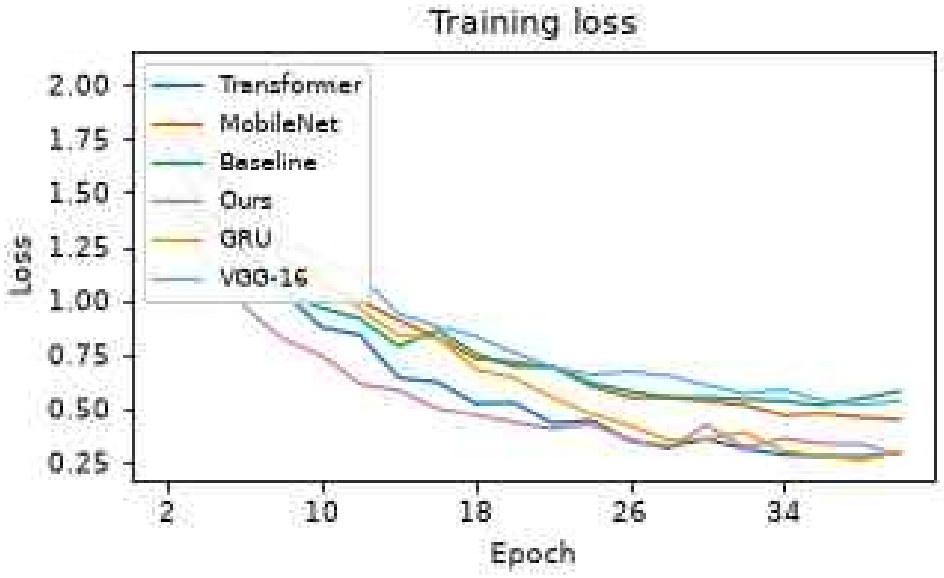


Figure 2: Training loss for the models in this experiment.

At epoch 32, the loss of Ours is 0.32 (Figure 2). Epoch 22 sees VGG-16 post a loss of 0.690.

3 Discussion

On SST-2, Random Forest obtains an accuracy of 83.5%. The differences between the models are discussed in the next section.